

Trust-aware resilience for connected vehicle platoons

From attack-aware state estimation
to an onboard edge module

Project overview for potential users and
decision-makers

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Connectivity creates physical risk

CONTEXT

Cooperative driving depends on data the vehicle cannot fully trust.

What connectivity enables

- Shared position, speed, heading and acceleration
- Cooperative cruise control and coordinated platoons
- Smoother traffic flow with fewer onboard measurements

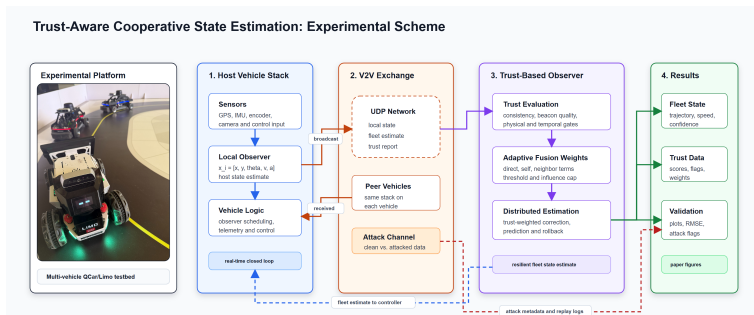
What an attack can change

- False, replayed, delayed or missing V2V packets
- Corrupted estimates that propagate across the fleet
- Unsafe control decisions based on plausible-looking data

Preserve cooperative value; degrade safely when V2V evidence is unreliable.

A trust layer protects cooperation

PROJECT CONCEPT



The trust layer screens, weighs and corrects data before the existing controller.

Board sits at the data-control boundary

ELECTRONIC BOARD INTEGRATION

Inputs

- IMU and local kinematics
- GNSS position and 1PPS timing
- Vehicle-bus data and V2V packets: Wi-Fi now, C-V2X later

Edge functions

- Time and packet-validity gates
- State estimation, trust and source weighting
- Rollback, re-anchoring and audit logs

Outputs

- Trusted state estimates
- Attack, trust, availability and health flags
- CAN/CAN-FD or gateway data for controller and planner

Integration principle: start in monitoring mode; introduce control authority only after timing, robustness and safety gates are demonstrated.

Intervention is graduated, not binary

RESPONSE WHEN AN ATTACK IS DETECTED

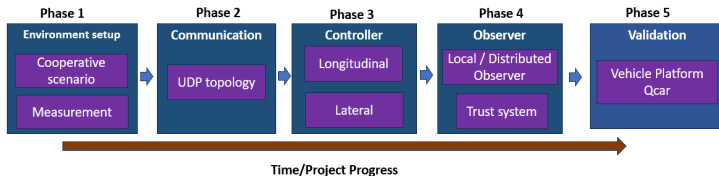


Two complementary effects: rejection stops future influence; rollback and anchoring revise recent contamination.

Current scope: the prototype does not claim direct actuator override; the vehicle controller retains command authority.

Validation moved from model to hardware

WORK PERFORMED



1. Numerical campaign

Five vehicles, controlled ground truth and multi-channel attacks.

2. Digital twin and scale models

QLabs + QCar/LIMO: packets, timing and closed-loop paths.

3. Embedded transfer

PCB: RTOS, sensors, radio, timing and logging.

Trust suppresses unreliable sources

APPROACH 1 - TRUST-AWARE DISTRIBUTED ESTIMATION

Reliability becomes an online fusion weight instead of a fixed assumption.

100%

Detection across all tested attacks and channel configurations

**80.64-
98.44%**

Attacked-source zero rate during the attack window

28-88 ms

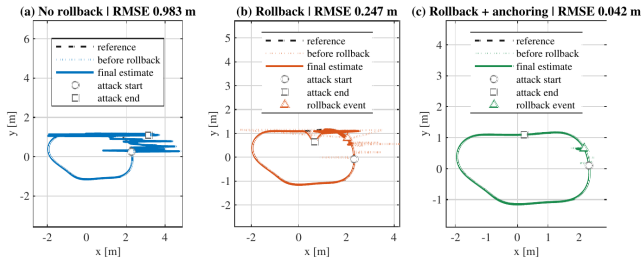
Mean threshold-crossing delay after the 10 s attack onset

Corrupted channel	Mean combined RMSE	Trust drop	Source zero rate
Local	0.643	0.319	80.64%
Global	0.393	0.460	92.90%
Local + global	0.567	0.544	98.44%

Rollback removes stored contamination

APPROACH 2 - DELAYED-DECISION RECOVERY

XY trajectory comparison for target V0



No rollback
0.983 m

Rollback
0.247 m

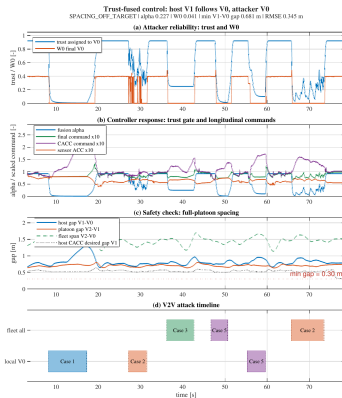
Rollback plus accepted
relative-pose anchoring
0.042 m

Local-channel Case 2 RMSE.

Correction reduces error 54.4%;
anchoring reduces it 75.2%.

Trust enables safe control degradation

APPROACH 3 - CONTROL INTERACTION



< 0.05

Mean attacked-source observer weight

0.681 m

V1 minimum path-projected gap

0.637 m

V2 minimum path-projected gap

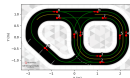
Both gaps remain above the study's 0.30 m reporting margin. The result is a feasibility check, not a formal full-footprint collision guarantee.

Tests reproduce the full operating loop

QUALITATIVE RESULTS



Physical QCar/LIMO scale-model track



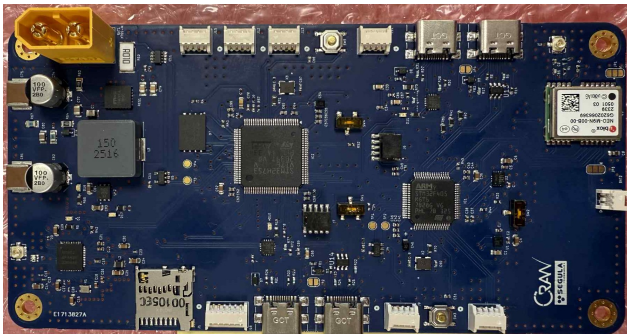
QLabs trajectory and waypoint scenario

One operating loop across both environments

Same state, communication and control concepts; real packet timing; repeatable attacks and logged recovery.

The deployment platform is tangible

MANUFACTURED ELECTRONIC BOARD



- Four-layer PCB with EMC-oriented grounding and power routing
- MCU/RTOS architecture targeting approximately 100 Hz estimation
- IMU, GNSS timing, Wi-Fi, storage and development interfaces

Bring-up evidence: stable RTOS operation at 100 Hz, UART, GNSS acquisition and functional Wi-Fi 6 tests.

Aligned with standards; not certified

POSITIONING VERSUS STATE OF THE ART

Transfer targets

- 3GPP NR V2X / C-V2X radio
- SAE J2735 and C-ITS interoperability
- ISO/SAE 21434 lifecycle risk engineering
- UNECE R155 CSMS and ISO 26262 safe degradation

Remaining industrialization

- Secure boot, signed updates, keys and HSM
- Automotive EMC and environmental qualification
- Production V2X, CAN gateway and diagnostics
- Threat analysis, safety case, false positives and stealth attacks

Research edge-estimation behind V2X - not a certified OBU or RSU.

Value comes from portable resilience

ECONOMIC RELEVANCE

01

Reuse across platforms

A portable edge node separates estimation and trust services from a specific vehicle computer or sensor stack.

02

Protect cooperation

Unreliable cooperative data can be isolated while local sensing and local control remain available.

03

Shorten evidence loops

Digital-twin reuse, configurable attacks and onboard logs accelerate testing and root-cause analysis.

Economic hypothesis for the pilot: lower integration effort and fewer attack-induced degradations than controller-only filtering or platform-specific implementations. Quantified ROI still requires vehicle-program data.

A shadow-mode pilot is the next gate

RECOMMENDED CONTINUATION

01

Connect production V2X and the target vehicle bus.

02

Run monitor-only; log trust, rejection and recovery.

03

Measure detection, false positives, timing and memory.

04

Authorize CACC-to-ACC fallback only after the gates pass.

Go / no-go evidence

- Timing / compute budget
- Detection and false positives
- Delay / dropout / stealth coverage
- EMC, power and interfaces
- Vehicle integration effort

Decision: pursue when a vehicle program provides bus access and a cooperative-driving use case.